# Kajol Dave

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## Summary

Data Scientist with 5 years of experience building statistical and machine learning models across credit risk, fraud detection, demand forecasting, and market intelligence. Hands-on track record of deploying production models, monitoring drift and performance, and translating outputs into business decisions for senior stakeholders across financial services, technology, and consumer markets.

## Education

**Master of Science in Business Analytics,** University of Massachusetts Amherst — Aug 2024 - May 2026
*Data Analytics Track* — (GPA: 4.0/4.0)
**Master of Business Administration,** Maharashtra Institute of Technology — July 2017 - May 2019
*International Business Track*
**Bachelor of Computer Application,** Savitribai Phule Pune University — July 2014 - May 2017

## Skills

**Languages and Tools:** Python (Pandas, NumPy, Scikit-learn), SQL, PySpark, dbt, MLflow, Airflow, Docker, AWS (SageMaker, S3, EC2, Redshift), Azure (ML, Synapse, Databricks), Snowflake, PostgreSQL, Git, Streamlit, FastAPI, Power BI, Tableau
**Modeling and Analytics:** Logistic Regression, Beta Regression, Classification, Ensemble Methods (XGBoost, LightGBM, Random Forest), Clustering, Time-Series Forecasting (ARIMA, ARIMAX, SARIMA, Prophet), SMOTE, Hyperparameter Tuning, Feature Engineering, SHAP, Bayesian Statistics, Cohort Analysis, Demand Forecasting, Scenario Modeling
**Credit Risk and Production ML:** PD/LGD/EAD Modeling, Scorecard Development, Stress Testing, Loss Forecasting, Vintage Analysis, Backtesting, Calibration, Model Governance, Model Risk Management, Real-Time Scoring, Drift Monitoring

## Experience

### Data Scientist - Bajaj Finance Ltd — Mar 2023 - Jun 2024

- Built Probability of Default models in Python with XGBoost and logistic regression scorecards on 500K+ loan records using credit bureau and internal loan data, lifting ROC-AUC from 0.76 to 0.82 and strengthening risk-tier separation across the portfolio
- Engineered utilization, and delinquency features in dbt on Redshift using WoE binning and IV-based selection, improving KS from 0.36 to 0.40 and lifting bad-capture in the top 3 deciles by 14% in champion-challenger evaluation against the incumbent scorecard
- Deployed containerized PD scoring models to AWS SageMaker for real-time scoring with model artifacts on S3 and dbt-managed feature pipelines populating a Redshift feature store, eliminating training-serving skew across batch and online environments
- Built Airflow-orchestrated monitoring pipelines tracking PSI-driven drift and AUC decay across 3 production models with automated retraining triggers, preventing silent drift between quarterly cycles and aligning with model risk management standards
- Calibrated decision thresholds via econometric loss models, expected-loss simulations, and vintage cohort stress testing, reducing projected default rates by 8% while holding monthly approval volumes within 1.5% of business targets
- Built supervised fraud classification model in Python and PySpark using XGBoost on 10M+ credit card transactions, applying class weighting on a 0.15% fraud rate to achieve 0.78 F1 and 92% recall at 1.5% false-positive rate via threshold optimization

### Data Scientist - Markets and Markets — May 2022 - Feb 2023

- Built ecosystem adjacency models in Python (Pandas, scikit-learn) mapping interconnected ICT segments across telecom, cloud, and infrastructure using Bloomberg Terminal, S&P CapIQ, and PitchBook data via AWS Redshift, identifying a $20M blockchain telecom adjacency for a Fortune 500 software client with $5M projected 2-year revenue
- Developed time-series forecasting models in Python (statsmodels, Prophet, scikit-learn) integrating macro indicators, technology adoption curves, and competitive intensity, achieving 10–15% MAPE across SD-WAN, cloud security, and 5G segments
- Designed scenario-modeling and macroeconomic stress-testing layers capturing supply constraints, hyperscaler capex shifts, and regulatory triggers, enabling stakeholders to stress-test investment theses across 5+ PE diligence engagements

### Data Analyst - Grand View Research — Feb 2021 - Apr 2022

- Built forecasting models (ARIMA, SARIMA, ARIMAX, Prophet) in Python and SQL on 50K+ records in Azure Synapse, validated via rolling-window backtesting, delivering digital infrastructure capex forecasts for Siemens Digital Industries
- Designed bottom-up market-sizing frameworks integrating Bloomberg Terminal, S&P CapIQ, PitchBook, and trade-flow data across 6 M&A diligence engagements for executive and PE stakeholders
- Automated ingestion and feature engineering of vendor disclosure data in Python and Azure Databricks with Databricks Jobs feeding Power BI dashboards, cutting manual research effort by 30% and freeing analyst capacity

### Junior Data Analyst - Zion Market Research — May 2019 - Feb 2021

- Cut reporting turnaround time by 20% by rebuilding Tableau dashboards tracking vendor share, M&A activity, and pricing dynamics, surfacing quarterly KPI shifts that informed stakeholder investment decisions
- Established backtesting and MAPE-tracking practices on forecasting deliverables, surfacing systematic forecast errors and contributing methodology revisions adopted across 10 published outputs
- Built bottom-up market-sizing models and segmentation in Python and SQL across 10+ industry verticals broken out by buyer persona, deployment type, and geography, surfacing subsegment growth pockets and informing client investment priorities

## Academic Projects

**LGD Modeling & Expected Loss Estimation** *(Python, AWS EC2, Docker, Streamlit, FastAPI, SHAP)*

- Built two-stage LGD model on 1M+ Freddie Mac records combining logistic regime classifier with Beta Regression and XGBoost ($R^2$ 0.25), estimating \$1.08B lifetime Expected Loss across \$246.82B UPB within PD × LGD × EAD framework
- Engineered DTI, LTV, and delinquency features with WoE binning and IV selection; validated via decile-level calibration, Brier decomposition, and vintage backtesting across 2019-2021 cohorts, with highest-risk segment carrying 39% of EL on 14% of loans
- Deployed two-stage model to AWS EC2 via Docker with MLflow tracking and Model Registry, GitHub Actions CI/CD, Streamlit, and FastAPI scoring endpoint with SHAP attribution, enabling stage-gated promotion on calibration thresholds

**Credit Card Fraud Detection System** *(Python, LightGBM, XGBoost, scikit-learn, SHAP, MLflow, DagsHub, Streamlit, Docker, Airflow, HuggingFace Spaces)*

- Developed a fraud detection pipeline on 590K transactions (28:1 class imbalance) in Python, achieving 0.58 PR-AUC and 0.92 ROC-AUC with LightGBM, benchmarked against 3 models under a time-based split to prevent temporal leakage
- Engineered 504 features including smoothed target encoding for high-cardinality fields, device-string parsing, and paired missingness flags, and produced SHAP global and local explanations for per-transaction reason codes and regulatory explainability
- Tracked 20+ experiments in MLflow on DagsHub with a model registry and PR-AUC promotion gate, and deployed a Dockerized Streamlit app on HuggingFace Spaces with Airflow DAGs for daily retraining and batch scoring